

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Electrical contact cleaner, Part Number 3824510
- Fuel System Clean Care Kit, Part Number 4919073
- Wire tie clamps, Part Number 3690499

Additional Service Items

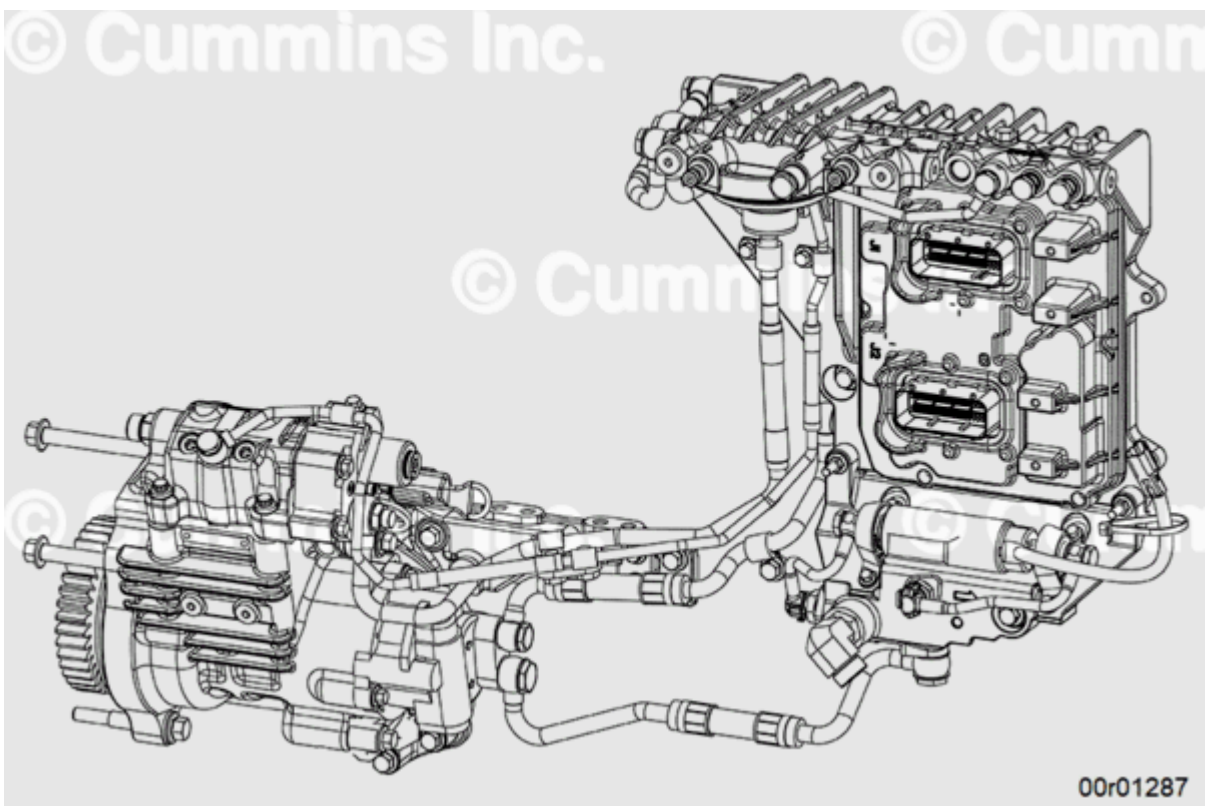
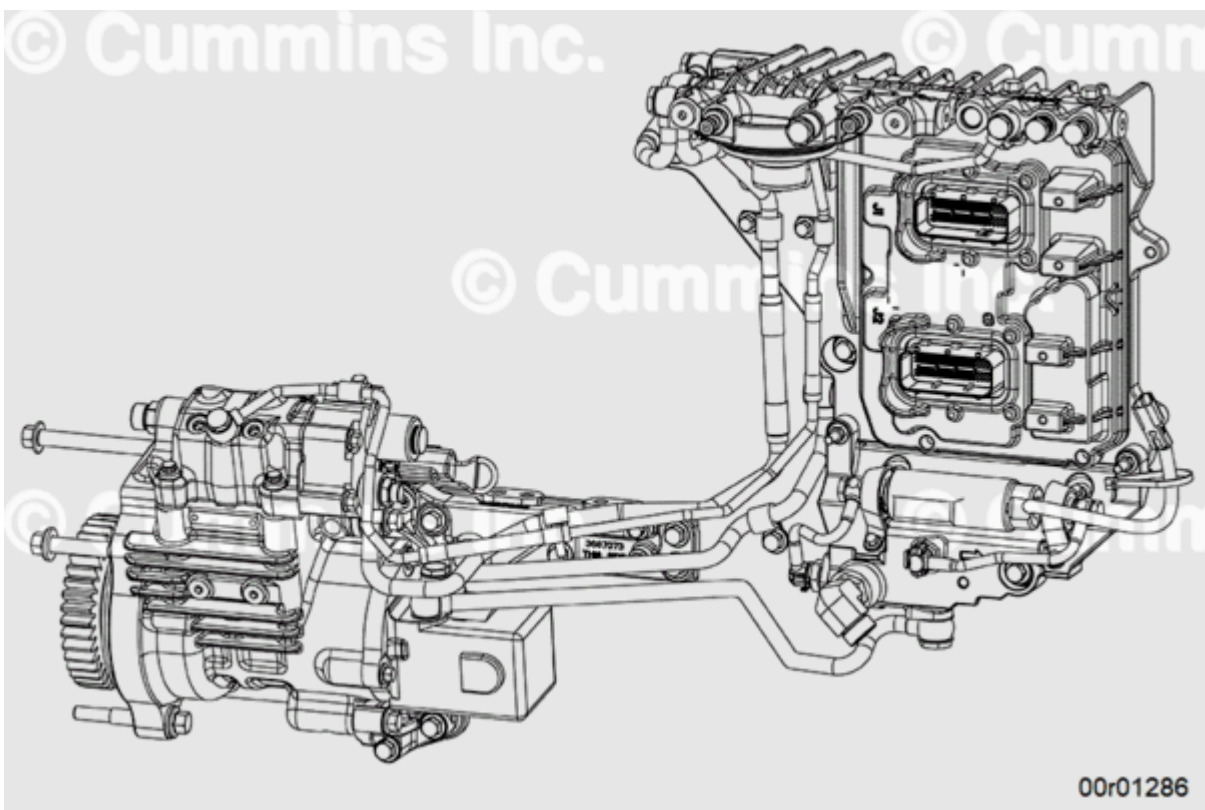
- No additional service items required.

General Information

Engines can be equipped with either standard or quick disconnect nylon fuel lines. Quick disconnect nylon fuel lines do **not** use the engine control module (ECM) cooling plate.

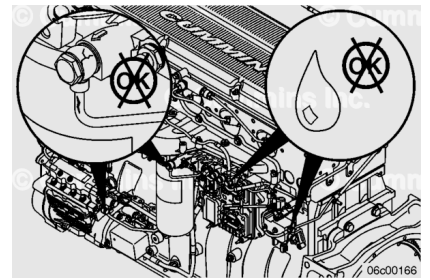
Engines can be equipped with two fuel pump assembly variations. These fuel pumps will have the same performance despite gear pump visual variation(s). Each fuel pump requires distinct fuel line configurations.

Cummins® Service Fuel Pumps:



Initial Check

Inspect all fuel supply lines and fittings. Look for cracks in the lines or leaking fittings.



Preparatory Steps

⚠ WARNING ⚠

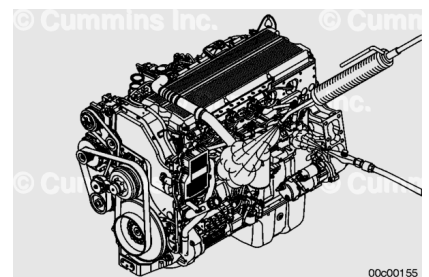
When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Clean all fittings before disassembly. Dirt or contaminants can damage the fuel system.



Before servicing **any** fuel system components, (such as fuel lines, fuel pump, injectors, etc.) which would expose the fuel system or internal engine component to potential contaminants prior to disassembly, clean the fittings, mounting hardware, and the area around the component to be removed. Dirt or contaminants can be introduced into the fuel system and engine if the surrounding areas are **not** cleaned, resulting in damage to the fuel system and engine. Refer to Procedure 000-009 in Section 0.

(/qs3/pubsys2/xml/en/procedures/99/99-000-009.html)

Clean the injector supply line connections and mating components with electrical contact cleaner, Part Number 3824510, or equivalent.

It is important to cover, cap, or plug any openings as soon as possible when servicing the fuel system in order to prevent engine damage. Caps and plugs can be found in Fuel System Clean Care Kit, Part Number 4919073.

⚠ WARNING ⚠

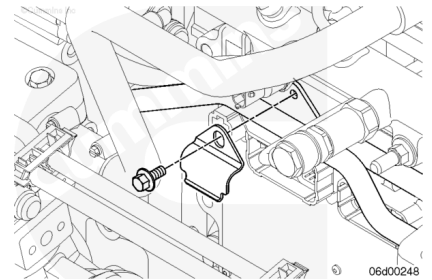
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.

Remove

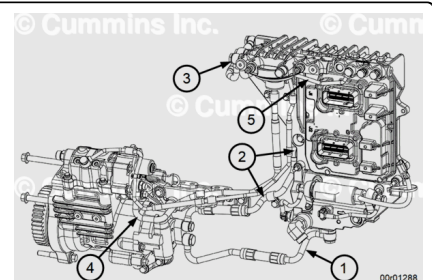
Without Quick Disconnect Fuel Lines

Remove the capscrews from the P-clips, as necessary.



⚠ CAUTION ⚠

Hold the banjo fittings while tightening the banjo bolt to prevent fitting rotation. Allowing the banjo fitting to rotate may damage the fuel line.



Fuel supply lines connect at the following joints:

1. Fuel lift pump support outlet to fuel pump gear pump inlet
2. Fuel pump gear pump outlet to ECM cooling plate inlet
3. ECM cooling plate outlet to fuel filter inlet
4. Fuel filter outlet to high-pressure fuel pump
5. Fuel filter outlet to aftertreatment fuel shutoff manifold.

Remove the fuel supply lines.

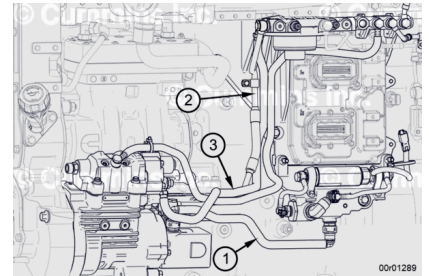
Discard the sealing washers.

With Quick Disconnect Fuel Lines

Fuel supply lines with quick disconnect fuel lines can be found at the following locations:

1. Fuel lift pump support outlet to fuel pump gear pump inlet
2. Fuel pump gear pump outlet to fuel filter inlet
3. Fuel filter outlet to high-pressure fuel pump.

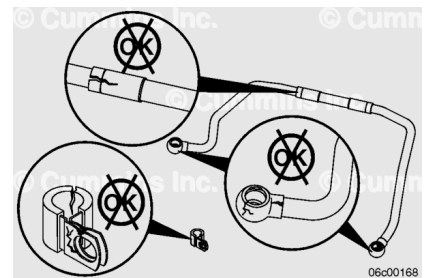
Remove the fuel supply lines.



Inspect for Reuse

Inspect for burrs or debris on banjo bolts, metal connectors, and quick disconnect nylon fuel lines, that can cause leaks.

Inspect the P-clips for damage. Replace any damaged P-clips.



Install

Without Quick Disconnect Fuel Lines

⚠ CAUTION ⚠

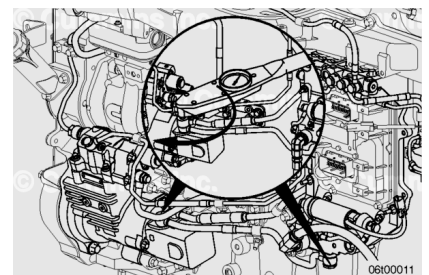
Hold the banjo fittings while tightening the banjo bolt to prevent fitting rotation. Allowing the banjo fitting to rotate may damage the fuel line.

Make sure the lines are routed and connected correctly. If the lines are connected incorrectly, the engine will **not** run.

Loosely install the fuel line banjos and P-clips.

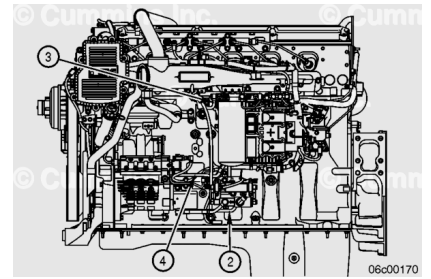
Tighten the fuel line connections.

- Fuel lift pump support to fuel pump gear pump inlet



Torque Value: 40 n•m [30 ft-lb]

- Fuel pump gear pump outlet to ECM cooling plate inlet (2)
- ECM cooling plate outlet to fuel filter inlet (3)
- Fuel filter outlet to high-pressure fuel pump (4)



Tighten the P-clip and banjo fittings.

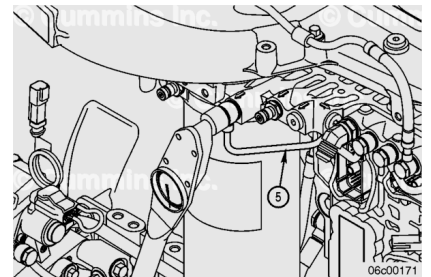
Torque Value:

M14 Banjo 34 n•m [25 ft-lb]

Torque Value:

P-clip 20 n•m [177 in-lb]

- Fuel filter outlet to aftertreatment fuel shutoff manifold (5)



Tighten the P-clip and banjo fittings.

Torque Value:

M14 Banjo 34 n•m [25 ft-lb]

Torque Value:

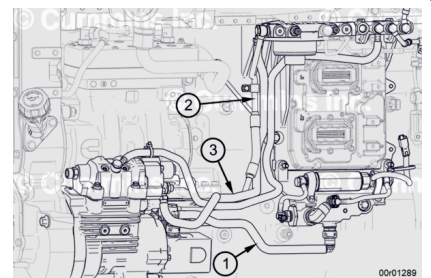
P-clip 20 n•m [177 in-lb]

With Quick Disconnect Fuel Lines

Make sure the lines are routed and connected correctly. If the lines are connected incorrectly, the engine will **not** run.

Loosely install the fuel line banjos and P-clips.

Tighten the fuel line connections and make sure to perform pull test on all quick connect fittings to make sure of a proper connection.



1. Fuel lift pump support outlet to fuel pump gear pump inlet
2. Fuel pump gear pump outlet to fuel filter inlet
3. Fuel filter outlet to high-pressure fuel pump

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the batteries. See equipment manufacturer service information.
- Prime the fuel system. Refer to Procedure 005-234 in Section 5.
(/qs3/pubsys2/xml/en/procedures/377/377-005-234-tr.html)
- Operate the engine. Check for leaks.

Last Modified: 26-May-2021
